ECO ARENA: Green Home Score & Retrofit Recommendation Engine

M1: Approach | Megathon X: Singularity | Problem Statement 2 (set by Lloyds)

1. Problem Statement and Our Understanding

Homeowners hold energy and property data but have no clear way to judge how efficient their home is, which improvements suit it, and what those improvements would change. Existing tools usually treat each retrofit in isolation and ignore interactions (e.g. insulation changes the benefit of a heat pump). The challenge asks for a system that answers four questions: **How green is my home today? What should I improve? How much could I save (energy per fuel, carbon)? Could it add property value?** We understand the task as building an explainable, reproducible decision system linking efficiency, energy, carbon, retrofit choice and property value, where all numbers are estimates, missing data is never fabricated, and any GenAI component must stay grounded in computed results.

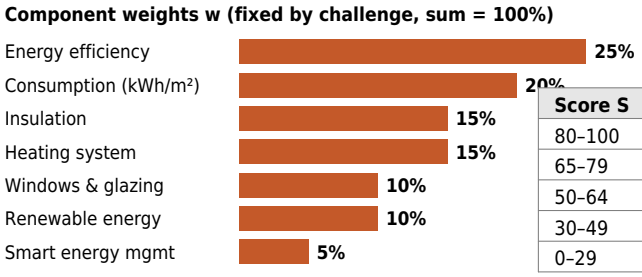
2. Idea / Abstract

ECO ARENA turns property and energy data into a personalised retrofit roadmap. It computes a deterministic **Green Home Score (0–100)**, flags Poor and Very poor homes, recommends feasible retrofits (insulation, heat pump, solar, double glazing, smart controls) ordered by their interactions, and quantifies per-fuel energy savings, carbon reduction and an indicative property-value range. Our key innovation is an **interaction-aware Retrofit Digital Twin (Retrofit Lab)**: the user selects one or several upgrades and the system re-runs the models (Current Home → Retrofit Scenario → New Home State), recalculating score, electricity, gas, carbon and value, so individual upgrades and combined packages can be compared. Guiding principle: **“Models calculate. The LLM explains.”** The LLM never computes or invents a score, saving, carbon or value figure.

3. High-Level Approach, Pipeline and Architecture

Pipeline: (1) Property, energy and sales data (synthetic + licensed public) → (2) Validation and feature engineering (unit harmonisation, kWh/m², fabric/heating/glazing/solar/smart flags; missing and conflicting values flagged) → (3) Green Home Score engine → (4) Efficiency detector → (5) Retrofit feasibility and recommendation engine → (6) Energy, carbon and property-value models → (7) What-if scenario engine → (8) Grounded LLM explanation layer (explain-only) → (9) Customer dashboard. Nodes 1–7 and 9 are deterministic calculation/display; only node 8 uses the LLM. Dashboard journey: Home Assessment → Green Home Dashboard → Retrofit Lab → Energy & Carbon Insights → Property Value → AI Home Advisor.

Green Home Score (deterministic): $S = \Sigma\, w_k \times s_k$, where s_k is a 0–100 sub-score from a documented mapping (e.g. kWh/m² thresholds, U-value/insulation level, heating type, glazing type, solar presence, smart controls) and w_k is the fixed weight shown in the chart ($\Sigma\, w_k = 1$). Example: sub-scores of 40 (efficiency), 30 (consumption), 50 (insulation), 20 (heating), 60 (glazing), 0 (renewables), 0 (smart) give $S = 0.25{\cdot}40 + 0.20{\cdot}30 + 0.15{\cdot}50 + 0.15{\cdot}20 + 0.10{\cdot}60 + 0 + 0 = 32.5$, band Poor. Every score is shown with its seven sub-scores for transparency. Missing components are flagged and lower confidence rather than being silently filled.



Score S	Band
80–100	Highly efficient
65–79	Good
50–64	Moderate
30–49	Poor
0–29	Very poor

4. Core Calculation Mechanisms (the working method)

Output	Formula / method	Notes
Improvement potential	100 × (E_base − E_after) / E_base	% of annual purchased energy (grid electricity + gas, kWh/yr) removable by feasible retrofits. Poor homes are ranked by this value. Reported as unavailable if baseline is zero/missing.
Energy saving, per fuel	ΔE_f = E_f,base − E_f,after (f = electricity, gas)	Gradient-boosted regressors estimate post-retrofit electricity and gas separately, per retrofit and for the package, net of interactions. An increase shows as a negative saving (e.g. heat pump: gas down, electricity up).
Carbon reduction	ΔC = ΔE_elec × EF_elec + ΔE_gas × EF_gas	EF in kg CO_2e/kWh from a documented source, stored with units. Positive ΔC means lower emissions. Net per-fuel result.
Property value	£ low to £ high	Same property predicted before and after retrofit, other traits held constant; accounts for location, size, type and market. Indicative range, not a guaranteed uplift.

Retrofit engine: feasibility rules per property remove unsuitable measures; remaining ones are scored on suitability, impact and dependencies and ordered by interaction (insulation before heat pump). Each recommendation states what, why it suits the property, energy and carbon impact, value relevance, priority (High/Medium/Low) and dependencies.
Retrofit Lab (what-if): user input → models re-run → new score, electricity, gas, carbon and value → LLM explains. Package savings are netted for interactions, not simply added.

Grounded GenAI (AI Home Advisor): the LLM receives only computed outputs and answers questions such as “Why is my score low?”, “Why was insulation first?”, “Why did electricity rise after a heat pump?”, “What if I combine solar and

insulation?". It reports "unavailable" when data is insufficient instead of guessing.

5. Data Strategy

Synthetic property, energy, retrofit and sales data generated by documented, physics-informed, seeded rules (per fuel, with retrofit interactions), plus licensed public datasets used within their terms. Emission factors are stored with units and sources. No real personal or confidential customer data is used. The Green Home Score stays deterministic because the weights are challenge-defined; machine learning is used where prediction adds value (post-retrofit energy per fuel, property value, feasibility). All scripts, label rules and run instructions are versioned and seeded so organisers can reproduce every output.

6. Planned Testing Methodology

General protocol: seeded train / validation / test splits; test homes never used for training or tuning; every model compared against a simple baseline; metrics reported on validation and held-out data, with the validation-to-test gap reported honestly; held-out predictions delivered in a fixed schema (property_id, green_home_score, score_band, improvement_potential_pct, retrofits_ranked, delta_elec_kwh, delta_gas_kwh, delta_co2e_kg, value_low_gbp, value_high_gbp).

Component	Test / metric	Baseline
Green Home Score	Unit tests: weights sum to 1, S in 0–100, band boundaries, missing-data handling	Hand-computed cases
Electricity / gas saving	MAE (kWh/yr), per fuel, incl. negative savings	Property-type mean
Improvement ranking	Spearman ρ vs reference ranking	Rank by floor size
Retrofit feasibility	F1 score	Majority class
Carbon	Check $\Delta C = \Sigma \Delta E_f \times EF_f$ against hand calculation; units and EF source verified	Manual calculation
Property value	MAPE and range coverage on held-out sales	Median £/m²
GenAI explanations	Groundedness ≥90%, factual accuracy ≥90%, unsupported claims <5% against computed outputs	Ungrounded LLM

Scenario tests: (a) missing or conflicting data is flagged and resolved by a documented rule, lowering confidence, never silently invented; (b) infeasible or interacting retrofits are filtered and netted, not added; (c) higher electricity after a heating change is shown as a negative electricity saving beside the gas saving; (d) too little information gives "unavailable" with the reason. **Responsible AI:** inputs, modelled estimates and unavailable information are clearly separated; fairness checked across property types, ages and regions; outputs labelled estimates with low-confidence cases flagged for human review.

Expected outcome: a transparent, explainable platform that tells homeowners and lenders how efficient a home is, what to improve first, what it could save in energy and carbon, and what the home could become: **Measure → Diagnose → Prioritise → Simulate → Explain → Act.**